



Package Insert

KARDAM
Amlodipine Besilate
Tablets 5 mg and 10 mg
Rx Only

NAME OF DRUG PRODUCT:

Amlodipine Besilate Tablets 5 mg.
Amlodipine Besilate Tablets 10 mg.

(TRADE) NAME OF PRODUCT :

KARDAM 5.
KARDAM 10.

STRENGTH: 5 mg and 10 mg.

PHARMACEUTICAL DOSAGE FORM: Tablet.

ROUTE OF ADMINISTRATION: Oral

QUALITATIVE AND QUANTITATIVE COMPOSITIONS:

Amlodipine Besilate Tablets 5 mg

Each uncoated tablet contains Amlodipine Besilate equivalent to Amlodipine 5 mg.

Amlodipine Besilate Tablets 10 mg

Each uncoated tablet contains Amlodipine Besilate equivalent to Amlodipine 10 mg.

PHARMACEUTICAL FORM:

Amlodipine Besilate Tablets 5 mg: White to off white, flat, bevel edged, barrel shaped, uncoated tablets debossed with 'C' on one side and '58' on the other side.

Amlodipine Besilate Tablets 10 mg: White to off white, flat, bevel edged, round uncoated tablets debossed with 'C' on one side and '59' on the other side.

CLINICAL PARTICULARS:

Pharmacodynamic properties

Amlodipine besilate is a calcium ion influx inhibitor of the dihydropyridine group (slow channel-blocker or calcium ion antagonist) and inhibits the transmembrane influx of calcium ions into the cardiac and vascular smooth muscle.

The mechanism of the antihypertensive action of Amlodipine besilate is due to a direct relaxant effect on vascular smooth muscle. Amlodipine besilate reduces total ischemic burden by the following two actions:

- 1) Amlodipine besilate dilates peripheral arterioles and thus, reduces the total peripheral resistance (after load) against which the heart works. Since the heart rate remains stable, this unloading of the heart reduces myocardial energy consumption and oxygen requirements.
- 2) The mechanism of action of Amlodipine besilate also probably involves dilatation of the main coronary arteries and coronary arterioles, both in normal and ischaemic regions. This dilatation increases myocardial oxygen delivery in patients with coronary artery spasm (Prinzmetal's or variant angina).

In patients with hypertension, once daily dosing provides clinically significant reductions of blood pressure in both the supine and standing positions throughout the 24 hour interval. Due to the slow onset of action, acute hypotension is not a feature of Amlodipine besilate administration.

In patients with angina, once daily administration of Amlodipine besilate increases total exercise time, time to angina onset and time to 1 mm ST segment depression and decreases both angina attack frequency and glyceryl trinitrate tablet consumption.

Amlodipine besilate is not associated with any adverse metabolic effects or changes in plasma lipids and is suitable for use in patients with asthma, diabetes and gout.

Pharmacokinetic properties:

Absorption, distribution, plasma protein binding: After oral administration of therapeutic doses, amlodipine is well absorbed with peak blood levels between 6-12 hours post-dose. Absolute bioavailability is estimated to be between 64 and 80 %. The volume of distribution is approximately 21 L/kg.

Biotransformation/elimination: The terminal plasma elimination half-life is about 35-50 hours and is consistent with once daily dosing. Amlodipine is extensively metabolized by the liver to inactive metabolites with 10 % of the parent compound and 60 % of metabolites excreted in the urine.

Use in the elderly: The time to reach peak plasma concentrations of amlodipine is similar in elderly and younger patients. Amlodipine clearance tends to be decreased with resulting increase in AUC and elimination half-life in elderly patients.

Indications:

First-line treatment of hypertension and as the sole agent to control blood pressure in the majority of patients. Patients not adequately controlled on a single antihypertensive agent may benefit from the addition of Amlodipine besilate, which has been used successfully in combination with a thiazide diuretic, β -adrenoceptor-blocking agent or an angiotensin-converting enzyme inhibitor.

Amlodipine besilate is indicated for the 1st-line treatment of myocardial ischemia, whether due to fixed obstruction (stable angina) and/or vasospasm/ vasoconstriction (Prinzmetal's or variant angina) of coronary vasculature. Amlodipine besilate may be used where a possible vasospastic/vasoconstrictive component but where vasospasm/vasoconstriction has not been confirmed. Amlodipine besilate may be used alone, as monotherapy or in combination with other antianginal drugs in patients with angina that is refractory to nitrates and/or adequate doses of beta-blockers.

Recommended Dose:

In adults: For both hypertension and angina, the usual initial dose is 5 mg Amlodipine besilate once daily, which may be increased to a maximum dose of 10 mg, depending on the individual patient's response.

No dose adjustment of Amlodipine besilate is required upon concomitant administration of thiazide diuretics, beta-blockers and angiotensin-converting enzyme inhibitors.

Use in children: Not recommended.

Use in the elderly: Amlodipine besilate, used at similar doses in elderly or younger patients, is equally well tolerated. Therefore normal dosage regimens are recommended.

Patient with renal impairment: Changes in amlodipine plasma concentrations are not correlated with degree of renal impairment, therefore the normal dosage is recommended. Amlodipine is not dialyzable.

Contraindications

Amlodipine besilate is contraindicated in patients with known sensitivity to dihydropyridines.

P1530307

A/s : 120 x 210 mm ■ Black

 AUROBINDO Packaging Development		Product Name	Component	Item Code	Date & Time
		Kardam	Leaflet	P1530307	13.07.2022 & 03:30 PM
		Customer / Country	Version No.	Reason Of Issue	Reviewed / Approved by
Malaysia	01	Revision			
Team Leader	Kiran K	Dimensions	No. of Colours : 01		
Initiator	Shirasha	120 x 210 mm	 30307		
Artist:	SCD	Pharmacode			
		30307			
Additional Information :		Supersede Item Code: P1516567	WOS-17-000986		
Sign / Date					

Warning and Precautions:

Renal Failure: Amlodipine is extensively metabolized to inactive metabolites with 10 % excreted as unchanged drug in the urine. Changes in amlodipine plasma concentrations are not correlated with degree of renal impairment.

Amlodipine may be used in such patients at normal doses. Amlodipine is not dialysable.

Impaired Hepatic Function: As with all calcium antagonists, amlodipine half-life is prolonged in patients with impaired liver function and dosage recommendations have not been established. The drug should therefore be administered with caution in these patients.

Use in the elderly: The time to reach peak plasma concentrations of amlodipine is similar in elderly and younger subjects. Amlodipine clearance tends to be decreased with resulting increases in AUC and elimination half-life in elderly. Amlodipine, used at similar doses in elderly or younger patients, is equally well tolerated. Therefore, normal dosage regimens are recommended.

Interaction with other Medicaments:

Amlodipine besilate has been safely administered with thiazide diuretics, beta blockers, angiotensin-converting enzyme inhibitors, long-acting nitrates, sublingual nitroglycerin, NSAIDs, antibiotics and oral hypoglycemic drugs.

The co-administration of amlodipine does not significantly alter the effect of warfarin on prothrombin response time.

The co-administration of amlodipine with digoxin did not change serum digoxin levels or digoxin renal clearance in normal volunteers, and that coadministration of cimetidine did not alter the pharmacokinetics of amlodipine.

The concomitant use of CYP3A4 inducers (e.g., rifampicin, hypericum perforatum) may give a lower plasma concentration of amlodipine. Amlodipine should be used with caution together with CYP3A4 inducers. Strong inhibitors of CYP3A4 (e.g., ketoconazole, itraconazole, ritonavir) may increase the plasma concentrations of amlodipine to a greater extent than diltiazem. Amlodipine should be used with caution together with CYP3A4 inhibitors.

Consumption of grapefruit/grapefruit juice should be avoided while taking Amlodipine. The intake of grapefruit juice may result in increased plasma amlodipine concentrations, which may enhance the blood pressure lowering effects of amlodipine. This interaction has been observed with other dihydropyridine calcium antagonists and represents a class effect.

Incompatibilities: Not relevant.

Pregnancy and Lactation

Amlodipine besilate should not be administered during pregnancy or lactation or to women of childbearing potential unless effective contraception is used.

Side Effects

Amlodipine is well tolerated. In patients with hypertension or angina, the most commonly observed side effects were headache, edema, fatigue, nausea, flushing, abdominal pain, palpitations and somnolence. Less commonly observed side effects in marketing experience include pruritus, rash, dyspnea, gingival hyperplasia, leucopenia, thrombocytopenia, hyperglycemia, Insomnia, mood changes hypertonia, hypoesthesia / paraesthesia peripheral neuropathy, syncope, taste perversion, tremor, visual disturbances, tinnitus, hypotension, vasculitis, cough, rhinitis, altered bowel habits, dry mouth, dyspepsia (including gastritis), pancreatitis, vomiting, alopecia, increased sweating, purpura, skin discoloration, urticaria, arthralgia, back pain, muscle cramps, myalgia, increased urinary frequency, micturition disorder, nocturia, Gynecomastia, impotence, asthenia, malaise, pain, weight increase/ decrease and rarely, erythema multiforme. As with other calcium-channel blockers, the following adverse events have been rarely reported and cannot be distinguished from the natural history of the underlying disease, eg myocardial infarction, arrhythmia (including ventricular tachycardia and atrial fibrillation) and chest pain.

Symptoms and Treatment of Overdose

The overdose causes an excessive peripheral vasodilation and possibly reflex tachycardia. Administration of activated charcoal immediately or up to two hours after ingestion of amlodipine 10 mg significantly decreases amlodipine absorption. Hypotension due to Amlodipine besilate overdose calls for active cardiovascular support including frequent monitoring of cardiac and respiratory function, elevation of extremities and attention to circulating fluid volume and urine output. A vasoconstrictor may be helpful in restoring vascular tone and blood pressure, provided that there is no contraindication to its use. Intravenous calcium gluconate may be beneficial in reversing the effects of calcium channel blockade. Since Amlodipine besilate is highly protein-bound, dialysis is not likely to be of benefit.

PHARMACEUTICAL PARTICULARS

List of Excipients

Microcrystalline cellulose, Sodium Starch Glycolate, Anhydrous Calcium Hydrogen Phosphate and Magnesium Stearate.

Incompatibilities

None known.

Shelf life

36 months.

Special precautions for storage

Store below 30°C.

Nature and contents of container

KARDAM 5 and KARDAM 10: 3 blisters of 10 tablets each.

MANUFACTURED BY:



AUROBINDO

Aurobindo Pharma Ltd., Unit-XV,

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Product Registration Holder In Malaysia

Healol Pharmaceuticals Sdn Bhd

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